THE SEARCH FOR A **LUNAR HOME** FOR

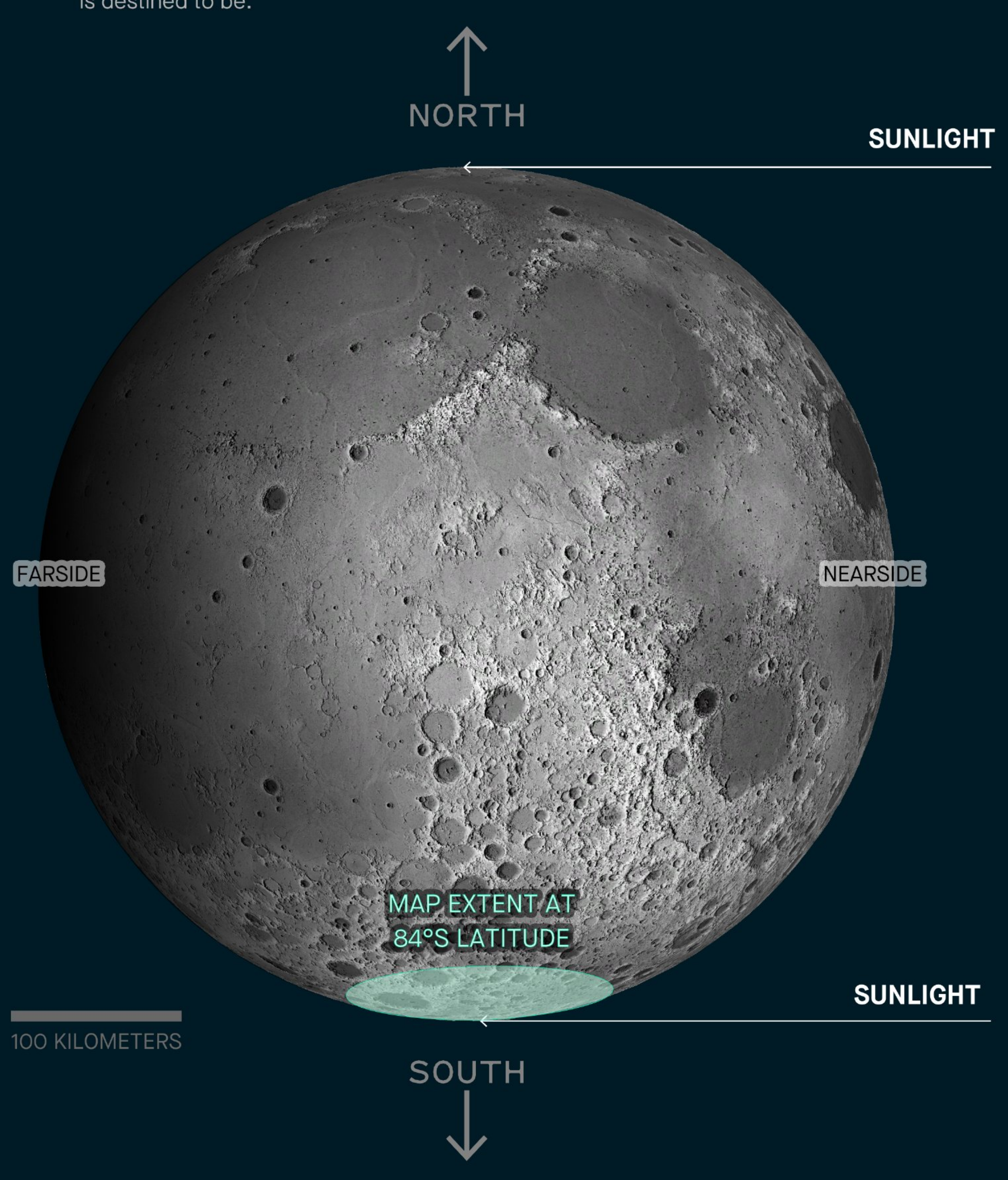
ASTRONAUTS

CARTOGRAPHY BY FILO MERID

BEFORE NASA SENDS ASTRONAUTS BACK TO THE MOON, has been scoping out the territory it has largely ignored since the last time it rocketed astronauts there over 50 years ago. Space agencies worldwide have spent the last decade shipping satellites strapping remote sensing technologies to map and measure the Moon in far greater detail than before.

Satellite data has provided convincing evidence of the presence of frozen water locked away from the presence of sunlight in what are known as permanently shadowed regions (PSRs) at the poles. Due to the Moon's relatively narrow tilt (1.5° vs Earth's 23.4°), pure water ice has been frozen at temperatures less than -300° for billions of years, unable to sublimate into outer space. Such evidence has been transformative as vital resources without the cost and burden of lifting heavy rocket fuel off the Earth, which has six times the gravity of the moon. With increasing interest has been focused on the South Pole where access to continuous sunlight has also been reliable at latitudes 84-90°. Such evidence has been transformative as vital resources without the cost and burden of lifting heavy rocket fuel and water off the Earth.

Further scrutiny is called for, however, as the wild, rugged terrain still has some mysteries to be accounted for. The maps displayed here offer a look at the data that has informed scientists in determining where exactly the perfect lunar home is destined to be.

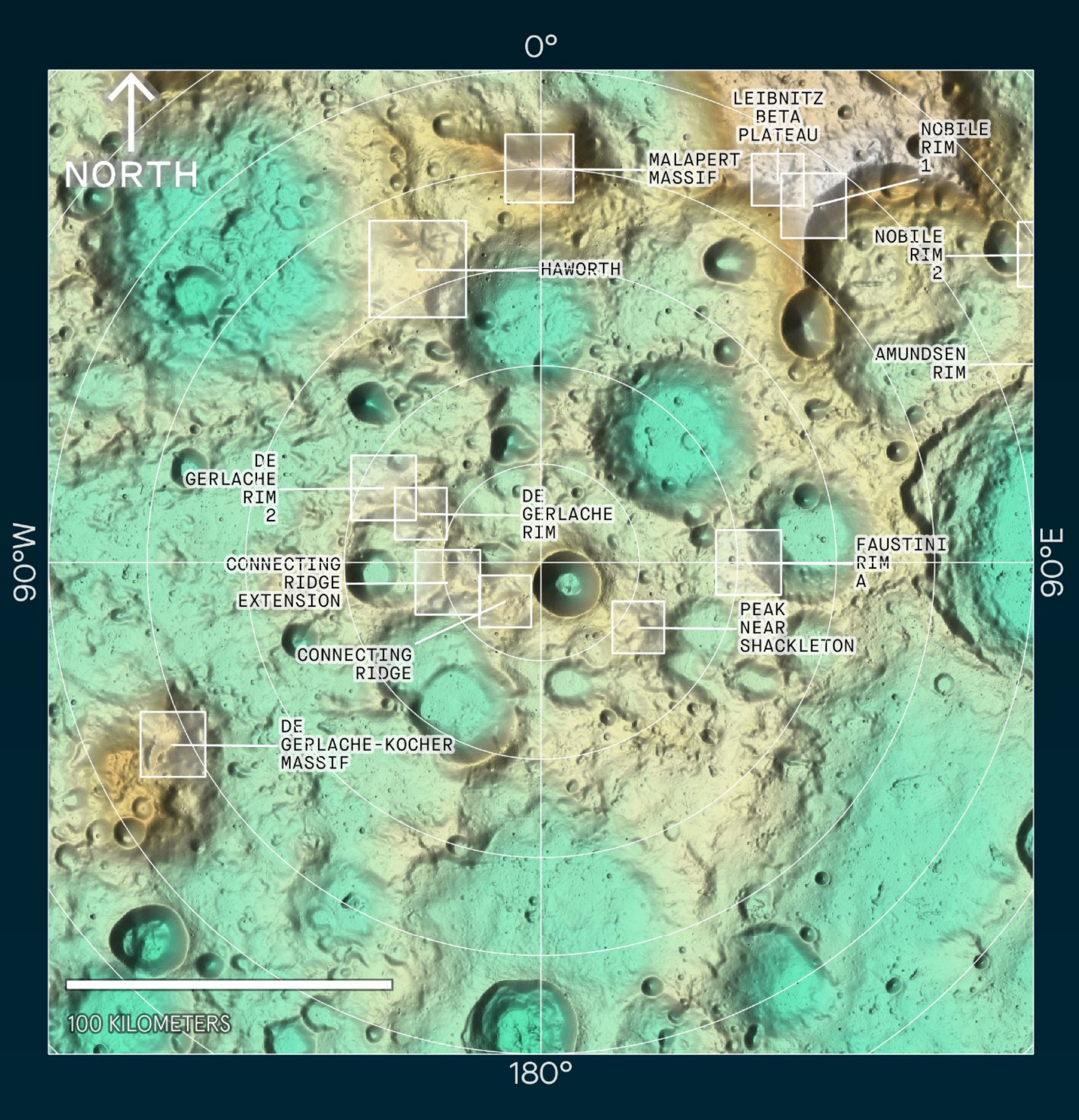


LUNAR TERRAIN & ELEVATION

Elevation (meters)

7010 -5480

Artemis Landing Sites ▨

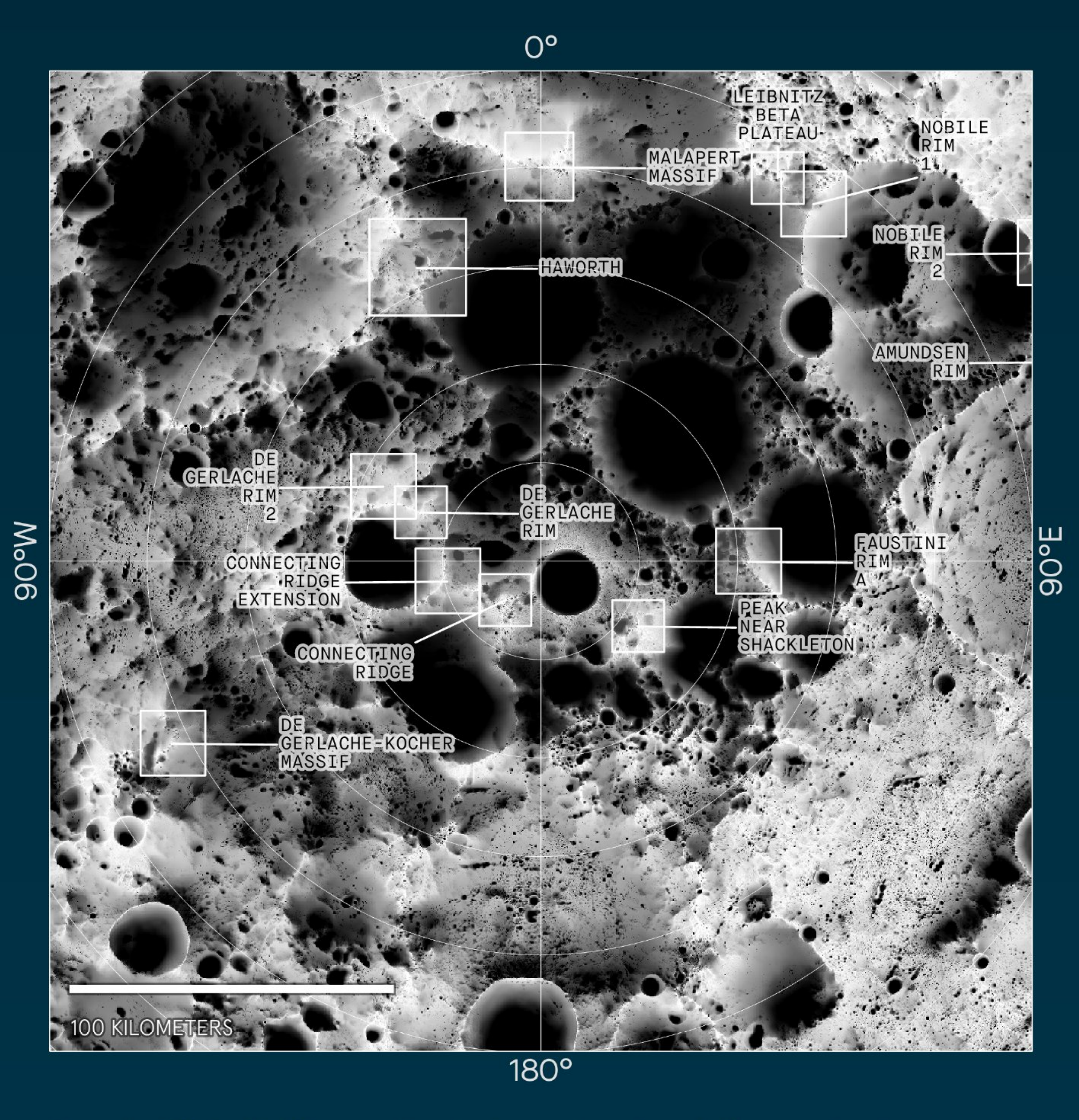


Comprehending lunar elevation at the South Pole is essential for ensuring secure spacecraft landings and astronaut well-being. The elevation data provides critical insights into the lunar topography, identifying height variations and potential hazards in landing areas. This information is pivotal for pinpointing suitable landing sites with flat and stable terrain, minimizing the risks of collisions or uneven descents. Precise elevation data is equally vital for assessing slopes, facilitating controlled spacecraft touchdowns on flat surfaces and reducing the likelihood of rough landings. This knowledge is indispensable for astronauts, aiding in planning extravehicular activities and traverses by anticipating potential obstacles and ensuring safe navigation across the lunar landscape. In essence, a comprehensive understanding of lunar elevation is foundational for mission planning, significantly contributing to the success and safety of lunar landings and subsequent surface operations.

AVERAGE SUN VISIBILITY

Visibility

HIGH LOW



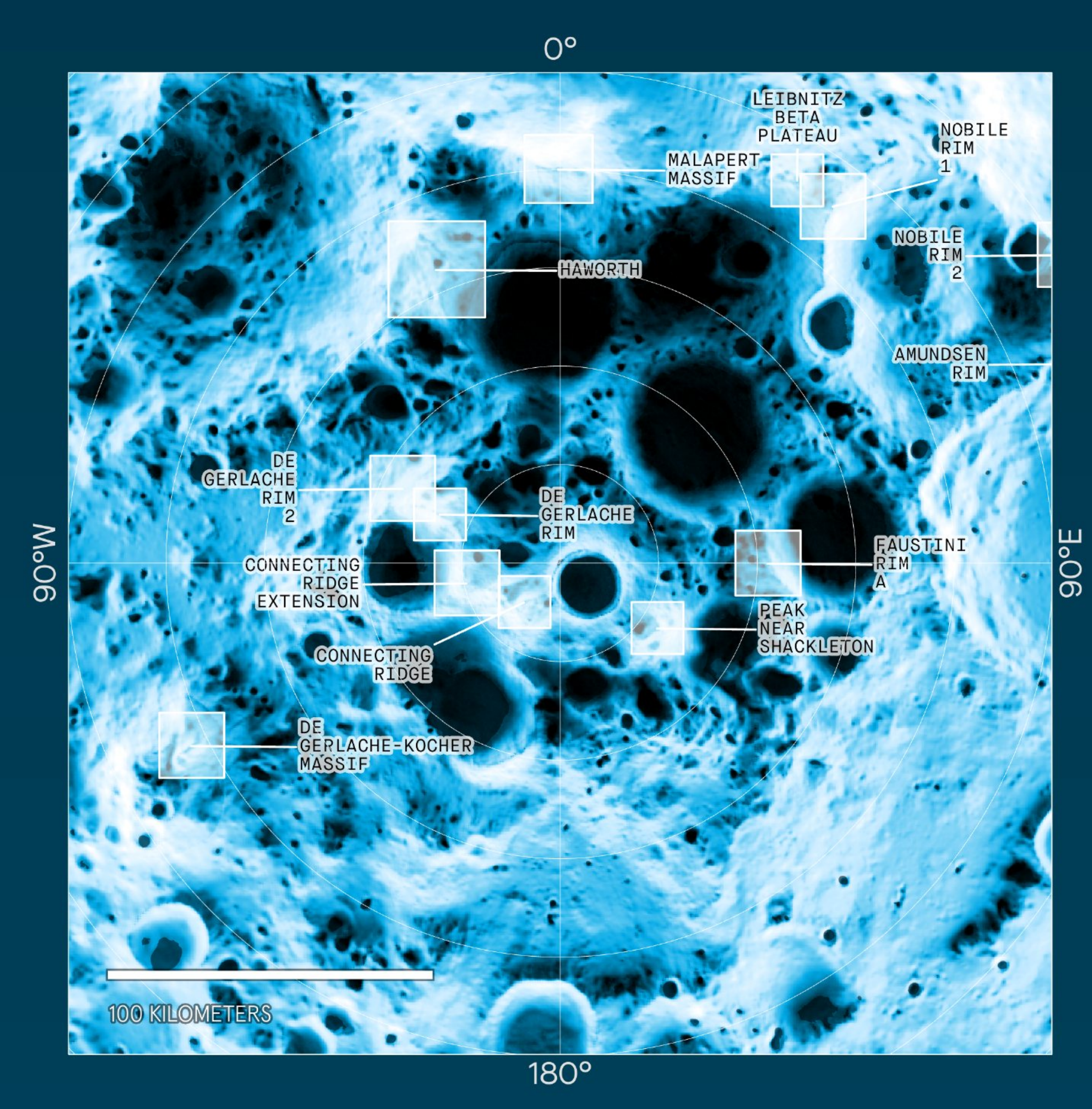
Due to the 1.5° tilt of the Moon, certain parts of the South Pole experience extended periods of sunlight due to their exposure to the sun almost continuously in addition to permanently shadowed regions. While this perpetual sunlight provides a valuable means of generating energy via solar panels on spacecraft and providing visibility for surface operations, continuous sunlight has a major drawback. Due to a lack of atmosphere, the Moons receives immense exposure to UV radiation, posing challenges for both equipment and astronaut safety.

Notably, NASA has determined landing sites that leverage sunlight visibility at any given time in the lunar day and year, though there is still debate on where exactly will give astronauts the least trouble in terms of long casted shadows.

AVERAGE TEMPERATURES

Temperature °F

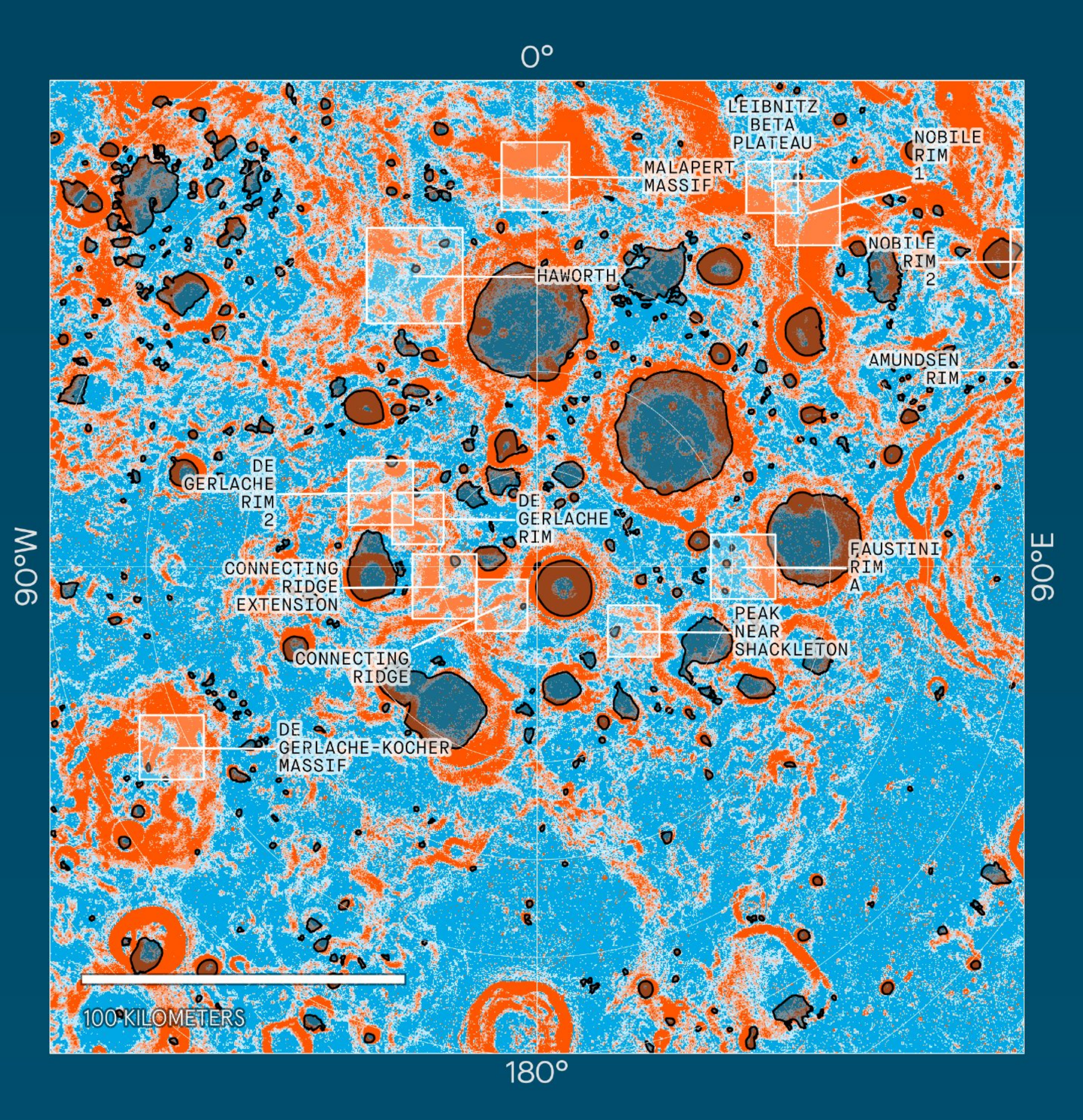
-77° -414°



Sunlight visibility at the lunar South Pole comes hand in hand with extreme temperature dynamics, ranging from intense cold during the lunar night to scorching heat in sunlight. The region, one of the coldest areas in our solar system, experiences temperatures between -77°F and -350°F. Efficient thermal management systems are vital for instrument functionality and survival, guiding the selection of landing sites that balance resource accessibility with thermal stability. Knowledge of temperature patterns is crucial for designing habitats that withstand the harsh lunar climate and insulate against extreme fluctuations, essential for optimizing resource utilization and harnessing solar energy. Additionally, temperature data aids in the search for water resources in permanently shadowed regions, where water ice may accumulate, contributing to sustaining human life and future lunar exploration. Despite the substantial benefits, operational challenges arise due to extreme conditions, necessitating sophisticated thermal control systems. Balancing these factors is crucial for planning successful and sustainable exploration endeavors at the lunar South Pole.

SLOPE & STABILITY

Slope (°)				PSR
29 - 86	14 - 21	21 - 29	0 - 14	▨



Understanding elevation and slope on the Moon is crucial for safe space operations and astronaut mobility. It is integral to identifying suitable landing sites, ensuring spacecraft touchdown on flat and stable terrain to minimize collision risks. This knowledge guides the selection of landing sites that balance resource accessibility with thermal stability, optimizing thermal management systems.

Additionally, this information is vital for astronauts planning extravehicular activities (EVAs) and traverses, as uneven terrain and steep slopes pose challenges, impacting mobility and safety during surface operations. Accurate elevation and slope data enable mission planners to anticipate obstacles, facilitating effective navigation for astronauts. Beyond mission planning, this knowledge contributes to a broader understanding of lunar geology and surface processes, enriching scientific endeavors and enhancing the success and safety of lunar missions.

SOURCES:
NASA (NATIONAL AERONAUTICS AND SPACE ADMINISTRATION), USGS (UNITED STATES GEOLOGICAL SURVEY), LROC HRSC (LUNAR RECONNAISANCE ORBITER HIGH-RESOLUTION STEREO CAMERA), UCLA DIVINER (UNIVERSITY OF CALIFORNIA, LOS ANGELES DIVINER LUNAR RADIOMETER EXPERIMENT), LUNAR & PLANETARY INSTITUTE, AND CALTECH (CALIFORNIA INSTITUTE OF TECHNOLOGY).

SPATIAL REFERENCES:
GCS: GCS MOON 2000
DATUM: MOON 2000
PROJECTION: STEREOGRAPHIC

NOTES: THE AZIMUTH FOR THESE MAPS IS NOT ACCURATE AS THE ANGLE OF THE SUN. THE MAXIMUM SOLAR ALTITUDE AT THE SOUTH POLE IS 1.5° AND IS HARSH. THE MAPS STRETCH THIS LIMIT TO 45° ABOVE THE HORIZON FOR A MORE CLEAR COMPREHENSION.